

Discrete Time Fourier Analysis

CHAPTER HIGHLIGHTS

- 🔊 Discrete Fourier Series
- 🔊 Properties of Discrete Time Fourier Series
- 🔊 Discrete Time Fourier Transform
- 🔊 Relation between Fourier Transform and z-transform
- 🔊 Properties of Discrete-time Fourier Transform
- 🔊 Common Fourier Transform Pairs
- 🔊 Frequency Response of Discrete-time LTI Systems
- 🔊 Discrete Fourier Transform (DFT)
- 🔊 Properties of DFT
- 🔊 The DFT as a Linear Transformation
- 🔊 Fast Fourier Transform (FFT)

DISCRETE FOURIER SERIES

Consider a periodic discrete-time signal $x(n) = x(n + N_0)$,
Where N_0 is fundamental period.

$$\text{Then, } x(n) = \sum_{k=0}^{N_0-1} C_k e^{jk\Omega_0 n}$$

$$\Omega_0 = \frac{2\pi}{N_0}$$

Where C_k are Fourier coefficients and are given as follows:

$$C_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x(n) e^{-jk\Omega_0 n}$$

Otherwise, it can be given as follows:

$$x(n) = \sum_{k=-\infty}^{\infty} C_k e^{jk\Omega_0 n}, \quad \Omega_0 = \frac{2\pi}{N_0}$$

$$C_K = \frac{1}{N_0} \sum_{n=-\infty}^{\infty} x(n) e^{-jk\Omega_0 n}$$

For $k = 0$,

$$C_0 = \frac{1}{N_0} \sum_{n=-\infty}^{\infty} x(n) \text{ is average value of } x(n) \text{ over a period.}$$

Fourier coefficients C_k are often referred to as the spectral coefficients of $x(n)$.

The Fourier coefficients C_k are complex, and they can be represented in polar form as $C_K = |C_K| e^{j\angle C_K}$

The plot of $|C_K|$ versus Ω is called amplitude spectrum and that of $\angle C_K$ versus Ω is called angle (or phase) spectrum. Both these plots are the frequency spectra of $x(n)$.

The spectrum of a signal $x(n)$, which is periodic with period N_0 , is a periodic sequence with period N_0 . Consequently, any N consecutive samples of the signal or its spectrum (C_K) provide a complete description of the signal in the time or frequency domains.

Discrete-time complex exponentials that differ in frequency by a multiple of 2π are identical.

$$\text{If } \phi_k(n) = e^{jk\Omega_0 n},$$

$$\text{then } \phi_{k+N_0}(n) = \phi_k(n)$$

$$e^{j(K+N_0)\Omega_0 n} = e^{j(K+N_0)\frac{2\pi}{N_0}n} = e^{jK\frac{2\pi}{N_0}n + j2\pi n} = e^{jK\frac{2\pi}{N_0}n} = e^{jK\Omega_0 n}$$

Since the sequences $e^{jk\Omega_0 n}$ are distinct only over a range of N_0 successive values of K , the summation (to represent $x(n)$ in terms of $e^{jk\Omega_0 n}$) need only include terms over this range.

Since the discrete-time Fourier series is a finite series, in contrast to the continuous time case, there is no convergence issues with discrete-time Fourier series.

PROPERTIES OF DISCRETE-TIME FOURIER SERIES

There are many similarities between the properties of discrete-time and continuous-time Fourier series.

If $x(n)$ and $y(n)$ are periodic signals with period N_0 and fundamental frequency $\Omega_0 = 2\pi/N_0$, then their Fourier series coefficients are C_K and D_K , respectively, which are also periodic with period N_0 . We can use the notation for Fourier series as follows:

$$x(n) \xrightarrow{\text{DFS}} C_K, y(n) \xrightarrow{\text{DFS}} D_K$$

1. Linearity

For the signal $Ax(n) + By(n)$, the Fourier series coefficients are $AC_K + BD_K$.

$$Ax(n) + By(n) \longleftrightarrow AC_K + BD_K$$

2. Time shifting

$$x(n - n_0) \longleftrightarrow C_K e^{-jk\Omega_0 n_0}, \quad \Omega_0 = \frac{2\pi}{N_0}$$

3. Frequency shifting

$$e^{jM\Omega_o n} x(n) \longleftrightarrow C_{K-M}$$

4. Conjugation

$$x^*(n) \longleftrightarrow C_{-k}^*$$

5. Time reversal

$$x(-n) \longleftrightarrow C_{-k}$$

6. Time scaling

$$x_m(n) = \begin{cases} x(n/m), & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m \end{cases}$$

If $x_m(n)$ is periodic with period on N_o , then its Fourier series coefficients are $\frac{1}{m}C_k$, and Fourier series coefficients are also periodic with period mN_o .

7. Periodic convolution

$$\sum_{r=\langle N \rangle} x(r) y(n-r) \longleftrightarrow N_o C_k D_k$$

Convolution of signals gives the multiplication of their Fourier series coefficients.

8. Multiplication

$$x(n) y(n) \longleftrightarrow \sum_{\ell=\langle N \rangle} C_\ell D_{k-\ell}$$

9. First difference

$$x(n) - x(n-1) \longleftrightarrow (1 - e^{-jk\Omega_o}) C_k$$

10. Running sum

$\sum_{k=-\infty}^n x(k)$ is finite valued and periodic only if $C_o = 0$, then

$$\sum_{k=-\infty}^n x(k) \longleftrightarrow \left(\frac{1}{1 - e^{-jk\Omega_o}} \right) C_k$$

11. Conjugate symmetry for real signals

If $x(n)$ is real, then $x(n) = x^*(n)$.

So, $C_K = C_{-K}^* [\because x^*(n) \longleftrightarrow C_{-K}^*]$.

$$= C_{N_o-K}^*$$

It implies that $\text{Re}\{C_K\} = \text{Re}\{C_{-K}\}$, $\text{Im}\{C_K\} = -\text{Im}\{C_{-K}\}$. Real part of C_K is an even function, and imaginary part of C_K is an odd function. Similarly, $|C_K| = |C_{-K}|$ and $\angle C_K = -\angle C_{-K}$. $|C_K|$ is even function, and $\angle C_K$ is odd function of K . Even and odd decomposition of real signals.

$x(n) = x_e(n) + x_o(n)$, then

$$x_e(n) \longleftrightarrow \text{Re}\{C_K\}$$

$$x_o(n) \longleftrightarrow j \text{Im}\{C_K\}$$

12. Periodicity of Fourier coefficient

$$C_K + N_o = C_K$$

13. Duality

$$x(n) \xrightarrow{\text{DFS}} C_K = C(k)$$

$$C(n) \xrightarrow{\text{DFS}} \frac{1}{N_o} x[-k]$$

14. If $x(n)$ is real

$$C_{-k} = C_{N_o-k} = C_K^*$$

Where C_K^* is complex conjugate of C_K .

15. If $x(n)$ is real and even, then its DFS coefficients are real and even.

While if $x(n)$ is real and odd, its DFS coefficients are imaginary and odd.

16. Parseval's theorem

$$\frac{1}{N_o} \sum_{n=\langle N_o \rangle} |x(n)|^2 = \sum_{k=\langle N_o \rangle} |C_K|^2$$

The Parseval's relation states that the average power in a periodic signal equals the sum of the average powers is all of its harmonic components.

Solved Examples**Example 1**

Find the Fourier coefficients for

$$x(n) = \sin(6\pi/5)n.$$

Solution

$x(n) = \sin(6\pi/5)n$ is periodic when $\frac{2\pi}{\omega_o}$ is

$$\text{Ratio of integers } \omega_o = \frac{2\pi M}{N}$$

$$\frac{6\pi}{5} = \frac{2\pi \times 3}{5}$$

So, $x(n)$ has a fundamental period of $N = 5$ here.

$$x(n) = \sin 3 \left(\frac{2\pi}{5} \right) n = \frac{1}{2j} \left[e^{j3 \left(\frac{2\pi}{5} \right) n} - e^{-j3 \left(\frac{2\pi}{5} \right) n} \right]$$

$x(n)$ is expressed as sum of exponentials.

By comparing with discrete Fourier series,

$$C_3 = \frac{1}{2j}, C_{-3} = \frac{-1}{2j}$$

and remaining coefficients over one period of length $N = 5$ are zero. Because of periodicity of coefficients $C_K = C_{K+N}$

$$C_3 = C_8 = C_{13} = \frac{1}{2j}, C_{-3} = C_2 = C_7 = C_{12} \dots \dots \frac{-1}{2j}$$

Example 2

Find the Fourier coefficients of the discrete signals.

(a) $x(n) = \cos \sqrt{3}\pi n$

(b) $x(n)$ is periodic with period $N = 4$ and $x(n) = \{1, 0, 0, 1\}$

Solution

- (a) $x(n) = \cos \sqrt{3} \pi n$, here $\omega_0 = \sqrt{3} \pi$, $\frac{2\pi}{\omega_0}$ is not a rational number, the signal is not periodic, so it cannot be expanded in Fourier series.
- (b) $x(n)$ is periodic with period 4, and $x(n) = \{1, 0, 0, 1\}$
We can write $x(n) = \{1, 0, 0, 1\}$

$$x(n) = \sum_{K=0}^{N-1} C_K e^{j(2\pi/N)Kn}$$

and Fourier coefficients

$$C_K = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-jK\left(\frac{2\pi}{N}\right)n}$$

Here, $N = 4$, $x(0) = 1$, $x(1) = x(2) = 0$,
 $x(3) = 1$

$$\begin{aligned} \text{So } C_K &= \frac{1}{N} \sum_{n=0}^3 x(n) e^{-jK\left(\frac{2\pi}{N}\right)n} \\ &= \frac{1}{4} \left(1 + 0 + 0 + 1 \cdot e^{-jK\frac{3\pi}{2}} \right) \\ C_K &= \frac{1}{4} \left(1 + e^{-jK\frac{3\pi}{2}} \right), K = 0, 1, 2, \text{ and } 3 \end{aligned}$$

$$C_0 = \frac{1}{2}, C_1 = \frac{1}{4}(1 + j), C_2 = 0, C_3 = \frac{1}{4}(1 - j)$$

Given signal is a real signal, from conjugate symmetry, we can check $C_K = C_{N-K}^*$

$$C_1 = C_{4-1}^* = C_3^*$$

DISCRETE-TIME FOURIER TRANSFORM

Similar to continuous time aperiodic energy signals, the frequency analysis of discrete-time aperiodic finite energy signals involves a Fourier transform of the time domain signal.

For a finite energy discrete time, aperiodic signal $x(n)$, the Fourier transform is defined as follows:

$$\begin{aligned} X(\Omega) &= F\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n} \\ X(n) &= F^{-1}\{X(\Omega)\} = \frac{1}{2\pi} \int_{2\pi} X(\Omega) e^{j\Omega n} d\Omega \end{aligned}$$

$x(n)$ and $X(\Omega)$ form a Fourier transform pair
 $x(n) \leftrightarrow X(\Omega)$

The Fourier transform, $X(\Omega)$, of $x(n)$ is in general complex function and is a continuous function of Ω .

$$X(\Omega) = |X(\Omega)| e^{j\phi(\Omega)}$$

$X(\Omega)$ is referred as the Fourier spectrum of $x(n)$, $|X(\Omega)|$ is called magnitude spectrum of $x(n)$, and $\phi(\Omega)$ is called phase spectrum of $x(n)$.

If $x(n)$ is real, $|X(\Omega)|$ is an even function, and $\phi(\Omega)$ is an odd function of Ω

Sufficient condition for the convergence of $X(\Omega)$ is

$$\sum_{n=-\infty}^{\infty} x(n) < \infty$$

The two basic differences between CTFT and DTFT are as follows:

- 1. Periodicity of Fourier transform:** For continuous-time signals, the Fourier transform, hence the spectrum of the signal, has frequency range of $(-\infty, \infty)$. In contrast, the frequency range for a discrete time signal is unique over the frequency interval of $(-\pi, \pi)$ or $(0, 2\pi)$. So, the Fourier transform of the signal will also be in the range $(-\pi, \pi)$ or $(0, 2\pi)$. The DTFT spectrum is periodic with period 2π .
- 2. Discrete nature of Fourier transform:** The signal is discrete in time, and the Fourier transform of the signal involves summation of terms. (instead of an integral as in the case of CTFT).

RELATION BETWEEN FOURIER TRANSFORM AND Z TRANSFORM

Fourier transform of $x(n)$, $X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$

$$\text{Z transform of } x(n), X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

If ROC of $X(z)$ contains the unit circle, then the Fourier transform $X(\Omega)$ of $x(n)$ equals $X(z)$ evaluated on the unit circle.

$$X(\Omega) = X(z)|_{z=e^{j\Omega}}$$

If $x(n)$ is absolutely summable, the Fourier transform of $x(n)$ can be obtained from the z transform of $x(n)$, with $z = e^{j\Omega}$, since ROC of $x(z)$ will contain the unit circle.

PROPERTIES OF DISCRETE-TIME FOURIER TRANSFORM**1. Periodicity**

Discrete-time Fourier transform (DTFT) is always periodic in Ω with period 2π .

$$X(\Omega + 2\pi) = X(\Omega)$$

2. Linearity

If $x_1(n) \longleftrightarrow X_1(\Omega)$ and $x_2(n) \longleftrightarrow X_2(\Omega)$, then
 $a_1 x_1(n) + a_2 x_2(n) \leftrightarrow a_1 X_1(\Omega) + a_2 X_2(\Omega)$.

3. Time shifting

If $x(n) \longleftrightarrow X(\Omega)$, then $x(n - n_0) \leftrightarrow e^{-j\Omega n_0} X(\Omega)$.

4. Frequency shifting

$$e^{j\Omega_0 n} x(n) \leftrightarrow X(\Omega - \Omega_0)$$

5. Conjugation

$$x^*(n) \leftrightarrow X^*(-\Omega)$$

6. Time Reversal

$$x(-n) \leftrightarrow X(-\Omega)$$

7. Time scaling

$$x_m(n) = \begin{cases} x(n/m) = x(k) & \text{if } n = km, k = \text{integer} \\ 0 & \text{if } n \neq km \end{cases}$$

$$x_m(n) \leftrightarrow X(m\Omega)$$

8. Duality

$$X(t) \xrightarrow{F.S} C_k = x[-k]$$

9. Differentiation in frequency

$$nx(n) \leftrightarrow j \frac{dX(\Omega)}{d\Omega}$$

10. Differencing

$$x(n) - x(n-1) \leftrightarrow (1 - e^{-j\Omega}) X(\Omega),$$

11. Accumulation

$$\sum_{k=-\infty}^{\infty} x(k) \leftrightarrow \pi X(0)\delta(\Omega) + \frac{1}{1 - e^{-j\Omega}} X(\Omega), |\Omega| \leq \pi$$

12. Convolution

$$x_1(n) \times x_2(n) \leftrightarrow X_1(\Omega) \cdot X_2(\Omega)$$

13. Multiplication

$$x_1(n) \cdot x_2(n) \leftrightarrow \frac{1}{2\pi} X_1(\Omega) \otimes X_2(\Omega)$$

Where $X_1(\Omega) \otimes X_2(\Omega) = \int_{2\pi} X_1(\theta) X_2(\Omega - \theta) d\theta$ is periodic convolution of $X_1(\Omega), X_2(\Omega)$.

14. Parseval's relation

$$\sum_{n=-\infty}^{\infty} x_1(n)x_2(n) = \frac{1}{2\pi} \int_{2\pi} X_1(\Omega) X_2(-\Omega) d\Omega$$

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{2\pi} |X(\Omega)|^2 d\Omega$$

Parseval's relation states that the total energy in the signal $x(n)$ can also be determined by integrating the energy per unit frequency, $|X(\Omega)|^2 / 2\pi$ over a full 2π interval of distinct discrete-time frequencies. $|X(\Omega)|^2$ is referred as the energy-density spectrum of the signal $x(n)$.

15. If $x(n)$ is real,

$$\text{Let } x(n) = x_e(n) + x_o(n)$$

Where $x_e(n)$ and $x_o(n)$ are even and odd components of $x(n)$

Let

$$x(n) \xrightarrow{DTFT} X(\Omega) = A(\Omega) + jB(\Omega)$$

$$= |X(\Omega)| e^{j\theta(\Omega)}$$

$$\text{Then } X(-\Omega) = X^*(\Omega), X(\Omega) = X^*(-\Omega)$$

$$x_e(n) \leftrightarrow \text{Re}\{X(\Omega)\} = A(\Omega)$$

$$x_o(n) \leftrightarrow j \text{Im}\{X(\Omega)\} = jB(\Omega)$$

We have

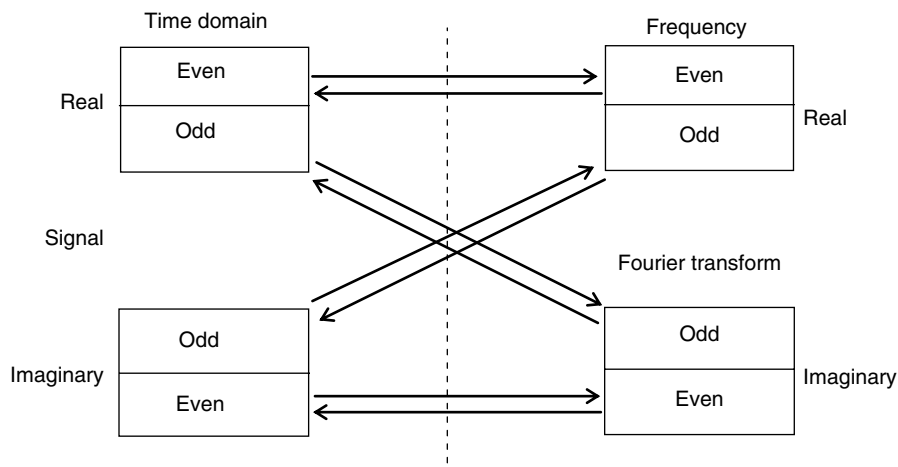
$$A(-\Omega) = A(\Omega), B(-\Omega) = -B(\Omega)$$

$$|X(-\Omega)| = |X(\Omega)| \quad \theta(-\Omega) = -\theta(\Omega)$$

If $x(n)$ is real and even, then $X(\Omega)$ is real and even, whereas if $x(n)$ is real and odd, $X(\Omega)$ is imaginary and odd.

Identical symmetry relationships hold for all Fourier representations.

Real and even time domain signals have real and even frequency domain representations, real and odd time domain signals have imaginary and odd frequency domain representations.



16. Correlation theorem

The cross correlation of two signals $x(n)$, $y(n)$ is

$$r_{xy}(m) = \sum_{n=-\infty}^{\infty} x(n) y(n-m),$$

$$m = 0, \pm 1, \pm 2, \dots$$

or
$$r_{xy}(m) = \sum_{n=-\infty}^{\infty} x(n+m) y(n),$$

$$m = 0, \pm 1, \pm 2, \dots$$

Then, the Fourier transform of $r_{xy}(m)$ is $S_{xy}(\omega)$.

$r_{xy}(m) \xrightarrow{DTFT} S_{xy}(\omega) = X(\omega) \cdot Y(-\omega)$ the function $S_{xy}(\omega)$ is called the cross energy density spectrum.

17. The Wiener–Khinchine theorem

Let $x(n)$ be a real signal, then autocorrelation sequence of $x(n)$ is

$$r_{xx}(l) = \sum_{n=-\infty}^{\infty} x(n) x(n-l)$$

Then, the Fourier transform of autocorrelation sequence is the energy spectral density of an energy signal.

$$r_{xx}(l) \xrightarrow{+} S_{xx}(\omega) = |X(\omega)|^2$$

18. The Fourier transform of a periodic signal

Consider a periodic sequence $x(n)$ with period N and the Fourier series representation

$$x(n) = \sum_{K=-\infty}^{\infty} C_K e^{jK(2\pi/N)n},$$

then, the Fourier transform of $x(n)$ is

$$X(\Omega) = \sum_{K=-\infty}^{+\infty} 2\pi C_K \delta\left(\Omega - \frac{2\pi}{N} K\right).$$

The Fourier transform of a periodic signal can be directly constructed from its Fourier coefficients

$$C_K = \frac{1}{N} \sum_{n=-\infty}^{\infty} x(n) e^{-jK(2\pi/N)n}$$

Example 3

Find the discrete-time Fourier transform of the signals.

(a) $x(n) = K^n u(n)$, $|K| < 1$

(b) $x(n) = K^{|n|}$, $|K| < 1$

Solution

(a) $x(n) = K^n u(n)$, $|K| < 1$

$$X(\Omega) = F\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$$

$$= \sum_{n=0}^{\infty} K^n e^{-j\Omega n} = \sum_{n=0}^{\infty} (K e^{-j\Omega})^n$$

$$X(\Omega) = \frac{1}{1 - K e^{-j\Omega}}$$

(b) $x(n) = K^{|n|}$, $|K| < 1$

We can write

$$x(n) = K^n u(n) + K^{-n} u(-n-1)$$

$$X(\Omega) = \sum_{n=0}^{\infty} K^n e^{-j\Omega n} + \sum_{n=-\infty}^{-1} K^{-n} e^{-j\Omega n}$$

$$= \sum_{n=0}^{\infty} (K e^{-j\Omega})^n + \sum_{m=1}^{\infty} (K e^{j\Omega})^m \text{ by keeping } m = -n$$

$$= \sum_{n=0}^{\infty} (K e^{-j\Omega})^n - 1 + \sum_{m=0}^{\infty} (K e^{j\Omega})^m$$

$$= \frac{1}{1 - K e^{-j\Omega}} - 1 + \frac{1}{1 - K e^{j\Omega}}$$

$$= \frac{1}{1 - K e^{-j\Omega}} + \frac{K e^{j\Omega}}{1 - K e^{j\Omega}}$$

$$= \frac{1 - K^2}{1 - 2K \cos \Omega + K^2}$$

Example 4

Consider the periodic signal $x(n) = \cos \omega_0 n$. Find the DTFT?

Solution

$$x(n) = \cos \omega_0 n$$

$$= \frac{1}{2} e^{j\omega_0 n} + \frac{1}{2} e^{-j\omega_0 n} \text{ is periodic with period.}$$

$$N = \frac{2\pi}{\omega_0} \Rightarrow \omega_0 = \frac{2\pi}{N}.$$

Then, its Fourier coefficients are C_{-1} and C_1 .

$$X(n) = \frac{1}{2} e^{j\left(\frac{2\pi}{N}\right)n} + \frac{1}{2} e^{-j\left(\frac{2\pi}{N}\right)n}$$

So, $C_{-1} = 1/2$, $C_1 = 1/2$

The Fourier transform

$$X(\Omega) = \sum_{K=-\infty}^{+\infty} 2\pi C_K \delta\left(\Omega - \frac{2\pi}{N} K\right)$$

$$= 2\pi \frac{1}{2} \delta\left(\Omega - \frac{2\pi}{N} (1)\right) + 2\pi \cdot \frac{1}{2} \delta\left(\Omega - \frac{2\pi}{N} (-1)\right)$$

$$= \pi [\delta(\Omega - \omega_0) + \delta(\Omega + \omega_0)].$$

COMMON FOURIER TRANSFORM PAIRS

$x(n)$	$X(\Omega)$
$\delta(n)$	1
$\delta(n - n_0)$	$e^{-j\Omega n_0}$
$x(n) = 1$	$2\pi \delta(\Omega), \Omega \leq \pi$
$e^{j\Omega_0 n}$	$2\pi \delta(\Omega - \Omega_0), \Omega_0 \leq \pi$

(Continued)

$\cos \Omega_0 n$	$\pi(\delta(\Omega - \Omega_0) + \delta(\Omega + \Omega_0))$, $ \Omega_0 \leq \pi$
$\sin \Omega_0 n$	$-j\pi(\delta(\Omega - \Omega_0) - \delta(\Omega + \Omega_0))$, $ \Omega_0 \leq \pi$
$u(n)$	$\pi\delta(\Omega) + \frac{1}{1 - e^{-j\Omega}}$, $ \Omega \leq \pi$
$-u(-n-1)$	$-\pi\delta(\Omega) + \frac{1}{1 - e^{-j\Omega}}$, $ \Omega \leq \pi$
$a^n u(n)$, $ a < 1$	$\frac{1}{1 - ae^{-j\Omega}}$
$-a^n u(-n-1)$, $ a > 1$	$\frac{1}{1 - ae^{-j\Omega}}$
$(n+1)a^n u(n)$, $ a < 1$	$\frac{1}{(1 - ae^{-j\Omega})^2}$
$a^{ n }$, $ a < 1$	$\frac{1 - a^2}{1 - 2a \cos \Omega + a^2}$
$x(n) = \begin{cases} 1 & n \leq N_1 \\ 0 & n > N_1 \end{cases}$	$\sin \left[\Omega \left(N_1 + \frac{1}{2} \right) \right] / \sin(\Omega / 2)$
$\frac{\sin \omega n}{\pi n}$, $0 < \omega < \pi$	$X(\Omega) = \begin{cases} 1 & 0 \leq \Omega \leq \omega \\ 0 & \omega < \Omega \leq \pi \end{cases}$
$\sum_{k=-\infty}^{\infty} \delta(n - kN_0)$	$\Omega_0 \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_0)$, $\Omega = \frac{2\pi}{N_0}$

FREQUENCY RESPONSE OF DISCRETE-TIME LTI SYSTEMS

The output $y(n)$ of discrete time LTI system equals the convolution of input $x(n)$ and impulse response $h(n)$.

So, $y(n) = x(n) * h(n)$

By taking Fourier transform

$$Y(\Omega) = X(\Omega) \cdot H(\Omega)$$

$$H(\Omega) = \frac{Y(\Omega)}{X(\Omega)}$$

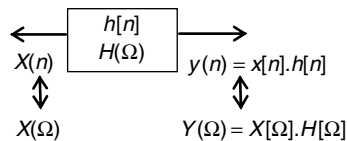


Figure 1 LTI discrete time system.

Let $H(\Omega) = |H(\Omega)| e^{j\theta H(\Omega)}$

The function $H(\Omega)$ is called frequency response, $|H(\Omega)|$ is magnitude spectrum, and $\theta H(\Omega)$ is phase spectrum of system.

$$Y(\Omega) = X(\Omega) \cdot H(\Omega)$$

In magnitude and phase representations, we can write it as follows:

$$|Y(\Omega)| = |H(\Omega)| \cdot |X(\Omega)|$$

$$\angle Y(\Omega) = \angle H(\Omega) + \angle X(\Omega).$$

The input magnitude is scaled by $|H(\Omega)|$ and is commonly known as gain of the system. The phase of the input $\angle X(\Omega)$ is modified by the LTI system by adding the phase $\angle H(\Omega)$ to it and is referred as the phase shift of the system.

Discrete-time LTI systems can be described by linear constant coefficient difference equations.

$$\sum_{k=0}^N a_k y(n-k) = \sum_{k=0}^M b_k x(n-k)$$

With $M \leq N$ by taking Fourier transform on both sides.

$$\sum_{k=0}^N a_k e^{-jk\Omega} y[\Omega] = \sum_{k=0}^M b_k e^{-jk\Omega} x[\Omega]$$

$$\text{Or } H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{\sum_{k=0}^M b_k e^{-jk\Omega}}{\sum_{k=0}^N a_k e^{-jk\Omega}}$$

This result is similar to the Z transform counterpart $H(Z) = Y(Z)/X(Z)$ with $Z = e^{j\Omega}$ and $H(\Omega) = H(\Omega + 2\pi)$

Periodic Nature of the Frequency Response

$$H(\Omega) = H(\Omega + 2\pi)$$

They unlike the frequency response of continuous time systems that of all discrete-time LTI systems is periodic with period 2π , and therefore, we need to observe the frequency response of a system only over the frequency range $0 \leq \Omega < 2\pi$ or $-\pi \leq \Omega < \pi$.

Distortionless Transmission

In several applications, we require that the output waveform be replica of the input waveform.

As in the continuous -time case, transmission is said to be distortionless, if input $x(n)$ and the output $y(n)$ satisfy the condition.

$$y(n) = K \cdot x(n - n_d)$$

Here n_d , the delay (in samples) is assumed to be integer.

The Fourier transform of the above equation yields

$$Y(\Omega) = KX(\Omega) e^{-j\Omega n_d}$$

$$Y(\Omega) = X(\Omega) H(\Omega)$$

So, $H(\Omega) = K e^{-j\Omega n_d}$ is the frequency response required for distortionless transmission.

$$|H(\Omega)| = K, \angle H(\Omega) = -\Omega n_d.$$

Hence, for distortion less transmission, the amplitude response $|H(\Omega)|$ must be constant, and phase response $\angle H(\Omega)$ must be linear function of Ω with slope $-n_d$.

For distortionless transmission, we require a linear phase characteristic. If the slope of $\angle H(\Omega)$ as a function of frequency (Ω) is constant, then it is a linear phase system. The slope can be expressed as $n_g(\Omega) = \frac{-d}{d\Omega} \angle H(\Omega)$

If $n_g(\Omega)$ is constant, all the components are delayed by n_g samples. However, if slope is not constant, the delay n_g varies with frequency. This variation means different frequency components undergo different amounts of delay, and consequently, the output is not replica of input waveform.

In case of LTI systems, $n_g(\Omega)$ is called as group delay or envelop delay.

$$\text{The phase delay } n_p(\Omega) = -\frac{\angle H(\Omega)}{\Omega}$$

Example 5

Consider the causal LTI system that is characterized by the difference equation. $y(n) - Ky(n-1) = x(n)$, $|K| < 1$, then the impulse response of the system is?

Solution

$$y(n) - Ky(n-1) = x(n),$$

By taking Fourier transform on both sides

$$Y(\Omega) - Ke^{-j\Omega} Y(\Omega) = X(\Omega),$$

the transfer function of the system

$$H(\Omega) = \frac{Y(\Omega)}{X(\Omega)} = \frac{1}{1 - Ke^{-j\Omega}} \quad |K| < 1.$$

By taking inverse Fourier transform, $h(n) = K^n u(n)$ is the impulse response of the system.

Example 6

Consider the causal LTI system that is characterized by the difference equation. $y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = 2x(n)$

and let the input to this system be $x(n) = \left(\frac{1}{4}\right)^n u(n)$, then find the impulse response of the system and output for the given input $x(n)$?

Solution

$$y(n) - 5/6y(n-1) + 1/6y(n-2) = 2x(n)$$

By taking Fourier transform on both sides

$$Y[\Omega] - \frac{5}{6}e^{-j\Omega} Y(\Omega) + \frac{1}{6}e^{-j2\Omega} Y(\Omega) = 2X[\Omega].$$

The transfer function $H(\Omega)$

$$\begin{aligned} &= \frac{Y(\Omega)}{X(\Omega)} = \frac{2}{1 - \frac{5}{6}e^{-j\Omega} + \frac{1}{6}e^{-j2\Omega}} \\ &= \frac{2}{\left(1 - \frac{1}{3}e^{-j\Omega}\right)\left(1 - \frac{1}{2}e^{-j\Omega}\right)} \end{aligned}$$

By using partial fractions

$$H(\Omega) = \frac{6}{1 - \frac{1}{2}e^{-j\Omega}} - \frac{4}{1 - \frac{1}{3}e^{-j\Omega}}$$

By taking inverse Fourier transform

$$h(n) = 6\left(\frac{1}{2}\right)^n u(n) - 4\left(\frac{1}{3}\right)^n u(n) \text{ is the impulse response}$$

of the system for input $x(n) = \left(\frac{1}{4}\right)^n u(n)$, $\xrightarrow{DTFT} X(\Omega)$

$$= \frac{1}{1 - \frac{1}{4}e^{-j\Omega}}$$

the output $y(n) = x(n) \times h(n)$.

$$Y(\Omega) = X(\Omega) \cdot H(\Omega)$$

$$= \frac{1}{\left(1 - \frac{1}{4}e^{-j\Omega}\right)} \cdot \frac{2}{\left(1 - \frac{1}{2}e^{-j\Omega}\right)\left(1 - \frac{1}{3}e^{-j\Omega}\right)}$$

By taking partial fractions $Y(\Omega) =$

$$\frac{16}{1 - \frac{1}{3}e^{-j\Omega}} - \frac{12}{1 - \frac{1}{2}e^{-j\Omega}} - \frac{6}{1 - \frac{1}{4}e^{-j\Omega}}$$

By taking inverse Fourier transform,

$$y(n) = 16\left(\frac{1}{3}\right)^n u(n) - 12\left(\frac{1}{2}\right)^n u(n) - 6\left(\frac{1}{4}\right)^n u(n)$$

is the output of the system for input $x(n)$

$$= \left(\frac{1}{4}\right)^n u(n).$$

DISCRETE FOURIER TRANSFORM

The technique known as the discrete Fourier transform (DFT) is used for finite length sequences.

Let $x(n)$ be a finite length sequence of length N , that is $x(n) = 0$ outside $0 \leq n \leq N-1$

$$\begin{aligned} \text{DFT of } x(n) \text{ is } X(k) &= \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi}{N}\right)kn} \\ k &= 0, 1, \dots, N-1 \end{aligned}$$

The inverse DFT is given by

$$\begin{aligned} x(n) &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{+j\left(\frac{2\pi}{N}\right)kn} \\ n &= 0, 1, \dots, N-1 \end{aligned}$$

The DFT and IDFT of a signal $x(n)$ are also periodic with period N , and it is sufficient to compute the results for only one period (0 to $N-1$)

$$\text{DFT: } X(K) = \sum_{n=0}^{N-1} x(n)W_N^{kn}$$

$$\text{IDFT: } x(n) = \frac{1}{N} \sum_{K=0}^{N-1} X(K)W_N^{-Kn}$$

$$\text{Where } W_N = e^{-j2\pi/N}, W_N^{Kn} = e^{-j\left(\frac{2\pi}{N}\right)Kn}$$

$$W_N^{-Kn} = e^{j\left(\frac{2\pi}{N}\right)Kn}$$

W_N is the N^{th} root of unity.

Periodicity property of phase factor W_N .

$$W_N^{K+N} = W_N^K$$

$$\text{Symmetry property: } W_N^{K+\frac{N}{2}} = -W_N^K$$

The choice of N is not fixed, if $x(n)$ has length $N_1 < N$. We want to assume that $x(n)$ has length N by simply adding $(N - N_1)$ samples with zero value. This addition of dummy sample is known as zero padding. Then, the resultant $x(n)$ is often referred as N -point sequence and $X(k)$ defined is referred as an N -point DFT.

By a proper choice of N , such as choosing it to be a power of 2 computational efficiencies can be gained.

Important Features of DFT

- (1) There is a one-to-one correspondence between $x(n)$ and $X(k)$
- (2) There is an extremely fast algorithm, called FFT for its calculation.
- (3) The DFT is closely related to the discrete Fourier series and the Fourier transform
- (4) The DFT is the appropriate Fourier representation for digital computer realization, because it is discrete and of finite length in both time and frequency domains.

Relation Between DFT and Other Transforms

The relationship between the DFT and the discrete Fourier series is $X(k) = NC_K$

The relation between Z transform and DFT is

$$X(k) = X(z) \Big|_{z=e^{j2\pi k/N}}, k=0, 1, \dots, N-1$$

$$X(z) = \frac{1-z^{-N}}{N} \cdot \sum_{K=0}^{N-1} \frac{X(k)}{1-e^{j2\pi k/N} z^{-1}}$$

The relationship between the DFT and the Fourier transform

$$X(k) = X(\Omega) \Big|_{\Omega=k2\pi/N} = X\left(\frac{k2\pi}{N}\right)$$

Therefore, $X(k)$ corresponds to the sampled $X(\Omega)$ at the uniformly spaced frequencies $\Omega = k2\pi/N$ for integer k .

Because of the relationship given previously, the properties, of DFT and DTFT are quite similar. Except that the

DFT $X(k)$ is a function of a discrete variable, while $X(\Omega)$ is of continuous variable.

The DFT variables n and k must be restricted to the range $0 \leq n, k < N$

The DFT shifts $x(n - n_0)$ or $X(k - k_0)$ imply $x(n - n_0)_{\text{mod } N}$ or $X(k - k_0)_{\text{mod } N}$, where the module notation $[m]_{\text{mod } N}$ means $[m]_{\text{mod } N} = m + iN$ for some integer i .

Such that $0 \leq [m]_{\text{mod } N} < N$

PROPERTIES OF DFT

1. Linearity

$$a_1x_1(n) + a_2x_2(n) \longleftrightarrow a_1X_1(k) + a_2X_2(k)$$

2. Duality: $X(n) \longleftrightarrow Nx(-k)$

3. Time shifting

$$x[n - n_0]_{\text{mod } N} \longleftrightarrow W_N^{Kn_0} X(k), W_N = e^{-j(2\pi/N)}$$

4. Frequency shifting

$$W_N^{-k_0n} x(n) \longleftrightarrow X(k - k_0)_{\text{mod } N}$$

5. Conjugated: $x^*(n) \longleftrightarrow X^*(-k)_{\text{mod } N}$

6. Time reversal: $x(-n)_{\text{mod } N} \longleftrightarrow X(-k)_{\text{mod } N}$

7. Symmetry properties of DFT

For an N -point sequence, $x(n)$ $0 \leq n \leq N-1$

$$x(n) \xrightarrow[N]{\text{DFT}} X(k)$$

$$x^*(n) \longleftrightarrow X^*(N - k)$$

$$x^*(N - n) \longleftrightarrow X^*(k)$$

For any real signal, $x(n) \longleftrightarrow X(k)$

$$X(k) = X^*(N - k), x(N - k) = X^*(k) = X(-k)$$

$$|X(N - k)| = |X(k)|$$

$$\angle X(N - k) = -\angle X(k) \text{ (or)}$$

$$\text{Re}\{X(k)\} = \text{Re}\{X(N - k)\}$$

$$\text{Im}\{X(k)\} = -\text{Im}\{X(N - k)\}$$

$$\frac{1}{2} (x(n) + x(N - n)) \longleftrightarrow \text{Re}\{X(k)\}$$

$$\frac{1}{2} (x(n) - x(N - n)) \longleftrightarrow j \text{Im}\{X(k)\}$$

$$\text{8. Parseval's theorem: } \sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X(k)|^2$$

9. Circular convolution

$$x_1(n) \otimes x_2(n) \leftrightarrow X_1(k) X_2(k)$$

$$x_1(n) \otimes x_2(n) = \sum_{i=0}^{N-1} x_1(i) x_2(n - i)_{\text{mod } N}$$

10. Multiplication

$$x_1(n) x_2(n) \leftrightarrow \frac{1}{N} X_1(k) \otimes X_2(k)$$

DFT AS A LINEAR TRANSFORMATION

We can view DFT and IDFT as linear transformation as sequence $\{x(n)\}$ and $\{X(k)\}$, respectively.

Let us define an N -point vector x_N of the signal sequence $x(n)$, $n = 0, 1, 2, \dots, N-1$, an N -point vector X_N of frequency samples and an $N \times N$ matrix W_N as

$$x_N = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}, \quad X_N = \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$$

$$W_N = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & W_N^1 & W_N^2 & \dots & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix}$$

The N -point DFT may be expressed in matrix form as $X_N = W_N x_N$.

Where W_N is the matrix of the linear transformation. If inverse exists for W_N , then

$$\text{IDFT: } x_N = W_N^{-1} X_N$$

$$\text{We can write } x_N = \frac{1}{N} W_N^* X_N$$

$$\text{We can conclude that } W_N^{-1} = \frac{1}{N} W_N^*$$

$$\text{So, } W_N W_N^* = I_N$$

Where I_N is an $N \times N$ identity matrix.

Example 7

Compute the DFT of the four-point sequence

$$x(n) = \{0, -1, 2, -3\}$$

Solution

The first step is to determine the matrix W_4 by using periodicity and symmetry property $W_N^{K+N} = W_N^K$, $W_N^{K+N/2} = -W_N^K$

The matrix W_4 can be expressed as follows:

$$W_4 = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & W_4^1 & W_4^2 & W_4^3 \\ 1 & W_4^2 & W_4^4 & W_4^6 \\ 1 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

$$X_4 = W_4 x_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 2 \\ -3 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 - 2j \\ 6 \\ -2 + 2j \end{bmatrix}$$

FAST FOURIER TRANSFORM (FFT)

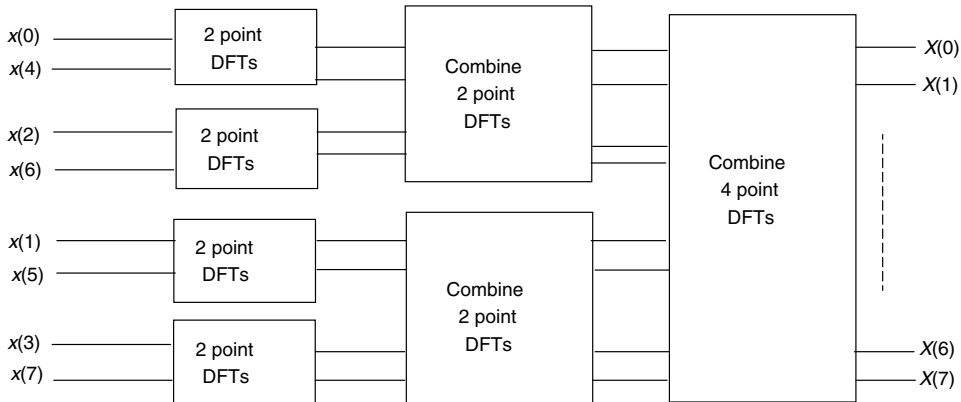
The Fast Fourier transform (FFT) is a discrete Fourier transform Algorithm which reduces the number of computations needed. The FFT is a simple method of laying out the computation, which is much faster for larger values of N , where N is the number of samples in the sequence.

The idea behind FFT is the divide and conquer approach, to break up original N point sample into two ($N/2$) sequences, this is because a series of smaller problems is easier to solve than one large one. FFT algorithms depend as the fact that $e^{-j\frac{2\pi}{N}}$ is a primitive root of unity, and thus can be applied to analogous transforms over any finite field.

FFT-Fast fourier transform algorithm reduce the problem of calculating an N point DFT to that of calculating many smaller-size DFTs. The computation is carried out separately on even-indexed and odd indexed samples to reduce the computational effort. FFT algorithm reduce storage requirements by performing computations in place by storing results in the same memory locations those previously held the data.

Three stages in the computation of an $N = 8$ -point DFT.

Evaluating DFT directly would take $O(N^2)$ arithmetic operations, FFT is an algorithm to compute the same result in only $O(N \log_2 N)$ operations.



EXERCISES

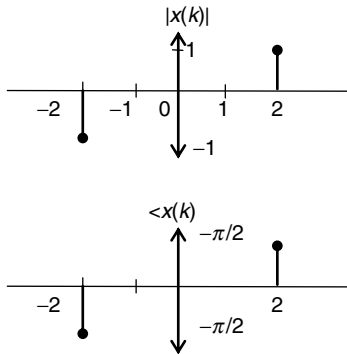
Practice Problems I

Direction for questions 1 to 15: Select the correct alternative from the given choices.

1. Determine discrete-time Fourier series coefficient $X(k)$ of the signal $x(n)$ as

$$x(n) = \cos\left(\frac{4\pi}{13}n\right) + \frac{1}{2}, \omega_0 = \frac{2\pi}{13}?$$

- (A) $\delta(k-1) + \delta(k) + \delta(k+2)$
 (B) $2(\delta(k-2) + \delta(k) + \delta(k+2))$
 (C) $\frac{1}{2}(\delta(k-2) + \delta(k) + \delta(k+2))$
 (D) $\frac{j}{2}(\delta(k-2) - \delta(k) + \delta(k+2))$
2. The discrete-time Fourier series coefficient of $x(n)$ is given as



Determine $x(n)$

- (A) $2j \sin\left(\frac{4\pi}{5}n + \frac{\pi}{2}\right)$ (B) $2 \sin\left(\frac{4\pi}{5}n + \frac{\pi}{2}\right)$
 (C) $2j \sin\left(\frac{\pi}{5}n + \frac{\pi}{2}\right)$ (D) $2 \cos\left(\frac{\pi}{5}n + \frac{\pi}{2}\right)$
3. Consider a discrete-time signal with Fourier representation $x(n) \xrightarrow{\text{DFTs: } \frac{\pi}{3}} X(k)$. The Fourier series coefficient $Y(k) = \cos\left(\frac{2\pi}{3}k\right)X(k)$. Determine corresponding signal $y(n)$?
 (A) $x(n-2) + x(n+2)$ (B) $x(n-2) - x(n+2)$
 (C) $\frac{1}{2}(x(n-2) + x(n+2))$ (D) $\frac{1}{2}x(n-2) - x(n+2)$
4. The 4-input discrete Fourier transform DFT of discrete time sequence $\{1, 0, 2, 3\}$ is
 (A) $\{6, -1+j3, 6, -1-j3\}$
 (B) $\{6, -1-j3, -6, -1+j3\}$
 (C) $\{6, -1-j3, 0, -1+j3\}$
 (D) $\{6, -1+j3, 0, -1-j3\}$
5. $\{x(n)\}$ is a real-valued periodic sequence with a period N , $x(n)$ and $X(k)$ form N -point discrete Fourier transform DFT pairs, the DFT $Y(k)$ of the sequence

$$y(n) = \frac{1}{N} \sum_{r=0}^{N-1} x(r)x(n+r) \text{ is}$$

- (A) $|X(k)|^2$ (B) $|X(k)|$
 (C) $|X(k)| - |X(k-x)|$ (D) None

6. A 5-point sequence $x(n)$ is given as $x[-3] = 1, x[-2] = 1, x[-1] = 0, x[0] = 5, x[1] = 1$, let $X(e^{j\omega})$ denote the discrete time Fourier transform of $x(n)$. The value of $\int_{-\pi}^{\pi} X(e^{j\omega})d\omega$ is
 (A) π (B) 5π (C) 10π (D) 20π
7. Let $x(n) = \left(\frac{1}{2}\right)^n u(n)$, $y(n) = x^2(n)$ and $Y(e^{j\omega})$ be the Fourier transform of $y(n)$, then $Y(e^{j0})$ is
 (A) $3/2$ (B) $2/3$ (C) $3/4$ (D) $4/3$
8. Determine the discrete-time Fourier coefficient $X(k)$ of the signal $x(n)$ as

$$x(n) = 4 \sin^2\left(\frac{\pi}{4}n\right)$$

- (A) $\delta(k) + \delta(k-1) - \delta(k+1)$
 (B) $2\delta(k) + \delta(k-1) + \delta(k+1)$
 (C) $2\delta(k) - \delta(k-1) - \delta(k+1)$
 (D) $\delta(k) - \delta(k-1) + \delta(k+1)$
9. Consider a discrete time signal with Fourier representation $x(n) \xrightarrow{\text{DFTs: } \frac{\pi}{5}} X(k)$. The Fourier series coefficient $Y(k) = X(k-2) - X(k+2)$, determine corresponding signal $y(n)$?
 (A) $2 \sin\left(\frac{2\pi}{5}n\right)x(n)$ (B) $j \sin\left(\frac{2\pi}{5}n\right)x(n)$
 (C) $2j \sin\left(\frac{2\pi}{5}n\right)x(n)$ (D) $\sin\left(\frac{2\pi}{5}n\right)x(n)$

Direction for questions 10 and 11:

Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x(n)$ where $x(n) = \{1, 0, 1, 2, -1, 0, -1\}$

10. Find $X(e^{j0})$
 (A) 2 (B) -2 (C) 4 (D) -4
11. Find $\int_{-\pi}^{\pi} X(e^{j\omega})d\omega$
 (A) 4π (B) -4π (C) $\pi/4$ (D) $-\pi/4$
12. The first six points of the 8-point DFT of a real-valued sequence are 4, $2-j3$, 0, $2+j$, 0 and $2-j$. The last two points of the DFT are, respectively.
 (A) 0, $2-j3$ (B) 0, $2+j3$
 (C) $2-j3, -4$ (D) $2+j3, -4$
13. Consider a length - 4 sequence given as $x(n) = \{1, -2, -1, 3\}$.
 The 4-point discrete Fourier transform of the sequence $x(n)$ is $X(k)$, then value of $X[2]$ is
 (A) 0 (B) 1 (C) 2 (D) -1
14. Each of two sequence $x(n)$ and $y(n)$ has a period $N = 4$, the Fourier coefficients are

$$C_0 = C_3 = \frac{1}{3}, C_2 = 1, C_1 = \frac{1}{2} \text{ and}$$

$-D_0 = D_1 = -D_2 = D_3 = 1$, respectively.

The Fourier coefficients for the signal $z(n) = x(n) y(n)$ will be

- (A) $\frac{-1}{2}, \frac{+1}{2}, -\frac{1}{2}, +\frac{1}{2}$ (B) $\frac{-1}{3}, \frac{+1}{3}, -\frac{1}{2}, +\frac{1}{3}$
 (C) $1, \frac{1}{2}, \frac{-1}{3}, \frac{+1}{3}$ (D) $\frac{-1}{2}, \frac{+1}{3}, \frac{+1}{2}, \frac{-1}{3}$

Practice Problems 2

Direction for questions 1 to 15: Select the correct alternative from the given choices.

1. The DFT of the sequence $x(n) = \{1, 1, -2, -2\}$

- (A) $1 + e^{-j\frac{\pi}{2}k} - 2e^{-j\pi k} - 2e^{-j\frac{3\pi}{2}k}$; for $k = 0, 1, 2, 3$
 (B) $1 - e^{-j\frac{\pi}{2}k} + 2e^{-j\pi k} - 2e^{-j\frac{3\pi}{2}k}$; for $k = 0, 1, 2, 3$
 (C) $1 - e^{+j\frac{\pi}{2}k} + 2e^{-j\pi k} + 2e^{-j\frac{3\pi}{2}k}$; for $k = 0, 1, 2, 3$
 (D) $1 + e^{-j\frac{\pi}{2}k} + 2e^{-j\pi k} + 2e^{-j\frac{3\pi}{2}k}$; for $k = 0, 1, 2, 3$

2. The inverse DFT of $Y(k) = \{1, 0, 1, 0\}$

- (A) $\{\frac{1}{2}, 0, \frac{1}{2}, 0\}$ (B) $\{\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}, 0\}$
 (C) $\{\frac{1}{\sqrt{2}}, 0, \frac{1}{2}, 0\}$ (D) $\{\frac{1}{2}, 0, \frac{1}{\sqrt{2}}, 0\}$

3. If DFT $\{x(n)\} = X(k)$, then DFT $\{x(n+m)\}_N$

- (A) $X(k) e^{j\frac{2\pi km}{N}}$ (B) $X(k) e^{j\frac{2\pi k}{mN}}$
 (C) $X(k) e^{-j\frac{2\pi k}{mN}}$ (D) $X(k) e^{-j\frac{2\pi k}{N}}$

4. In a N -point sequence, if $N = 16$, the number of complex additions and multiplications using Radix-2FFT are

- (A) 24, 12 (B) 64, 32 (C) 80, 64 (D) 64, 80

5. (1) DFT of even sequence is purely imaginary and DFT of odd sequence is purely real.

- (2) In DFT of real sequence, the real component is even and imaginary component is odd.

- (A) Both are true (B) Both are false
 (C) (1) false, (2) true (D) (1) true, (2) false

6. Let $x(n) = \left(\frac{2}{3}\right)^n u(n)$, $y(n) = x^2(n)$, and $Y(e^{j\omega})$ be the

Fourier transform of $y(n)$, then $Y(e^{j\omega})$ is

- (A) 4/3 (B) 4/9 (C) 4/5 (D) 9/5

7. The 4-point discrete Fourier transform (DFT) of a discrete-time sequence $\{1, -1, 0, 2\}$ is

- (A) $\{2, 1+3j, 0, 1-3j\}$ (B) $\{1+3j, 2, 1-3j, 0\}$
 (C) $\{0, 1+3j, 2, 1-3j\}$ (D) None of these

8. The discrete-time Fourier coefficients of $\cos\left(\frac{6\pi}{19}n + \frac{\pi}{4}\right)$ is

15. A real and odd periodic signal $x(n)$ has fundamental period $N = 5$, and Fourier series coefficients are $C_{11} = j$, $C_{12} = -j$, $C_{13} = 2j$. Then, the values of Fourier coefficients $C_0, C_{-1}, C_{-2}, C_{-3}$ respectively

- (A) $0, j, -j, 2j$ (B) $0, -j, j, -2j$
 (C) $2j, j, -j, -2j$ (D) $+2j, -j, j, -2j$

- (A) $\frac{1}{2}e^{-j\frac{\pi}{4}}, \frac{1}{2}e^{j\frac{\pi}{4}}$ (B) $\frac{1}{19}e^{-j\frac{\pi}{4}}, \frac{1}{19}e^{j\frac{\pi}{4}}$

- (C) $\frac{2}{19}e^{-j\frac{6\pi}{19}}, \frac{2}{19}e^{j\frac{6\pi}{19}}$ (D) $\frac{1}{19}e^{-j\frac{6\pi}{19}}, \frac{1}{19}e^{-j\frac{6\pi}{19}}$

9. The discrete Fourier transform of a sequence $x(n)$ is $X(k) = \{0, 1, 2, 3\}$.

The DFT of $y(n) = x(n-2)$ is.

- (A) $\{0, 1, 2, 3\}$ (B) $\{0, -1, 2, -3\}$
 (C) $\{0, 1, -2, 3\}$ (D) $\{2, 3, 0, 1\}$

10. The Fourier coefficients of $Y(k)$ of the signal $y(n) = |(-1)^n x(n)|$ are, ($N = 2m$, N, m are integers, N is period of $x(n)$)

- (A) $X\left[k - \frac{N}{2}\right]$ (B) $X[k - N]$

- (C) $X\left[k + \frac{N}{2}\right]$ (D) $X[k + N]$

11. The Fourier representation of a discrete and periodic signal is

- (A) continuous and a periodic
 (B) discrete and a periodic
 (C) continuous and periodic
 (D) discrete and periodic

12. $x(n)$ is a real periodic sequence, with period N and Fourier coefficient of $x(n)$ is given as $C_k = a_k + jb_k$, a_k, b_k are real, then

- (A) $a_{-k} = -a_k, b_{-k} = b_k$ (B) $a_{-k} = a_k, b_{-k} = -b_k$
 (C) $a_{-k} = -a_k, b_{-k} = -b_k$ (D) $a_{-k} = a_k, b_{-k} = b_k$

13. The relationship between the DFT and the discrete Fourier series is?

- (A) $X(k) = NC_k$ (B) $\frac{1}{N} X(k) = C_k$

- (C) $X(k) = \sum_{\ell=-\infty}^{\infty} C_{k-\ell}$ (D) $X(k) = C_k$

14. Which of the following is the discrete time Fourier transform of $(n+1)a^n u(n)$ (when $|a| < 1$)?

- (A) $\frac{1}{(1-ae^{j\Omega})^2}$ (B) $\frac{2a}{(1-ae^{j\Omega})^2}$

- (C) $\frac{2a}{(1-ae^{-j\Omega})^2}$ (D) $\frac{1}{(1-ae^{-j\Omega})^2}$

15. Two parallel connected discrete-time system with impulse responses $h_1(n)$, $h_2(n)$. The overall system impulse response is

- (A) $h_1(n) + h_2(n)$ (B) $h_1(n) h_2(n)$
 (C) $\frac{h_1(n)h_2(n)}{h_1(n) + h_2(n)}$ (D) $h_1(n) \times h_2(n)$

PREVIOUS YEARS' QUESTIONS

1. Consider the sequence

$$X(n) = [-4 - j5, 1 + j2, 4]$$

The conjugate anti-symmetric part of the sequence is [2004]

- (A) $[-4 \quad -j2.5 \quad j2 \quad 4 - j2.5]$
 (B) $[-j2.5 \quad 1 \quad j2.5]$
 (C) $[-j5 \quad j2 \quad 0]$
 (D) $[-4 \quad 1 \quad 4]$

2. Let

$x(n) = \left(\frac{1}{2}\right)^n u(n)$, $y(n) = x^2(n)$ and $Y(e^{j\omega})$ be the Fourier transform of $y(n)$, then $Y(e^{j0})$ is [2005]

- (A) $\frac{1}{4}$ (B) 2
 (C) 4 (D) $\frac{4}{3}$

3. Match the following and choose the correct combination: [2005]

	Group 1	Group 2
E	Continuous and aperiodic signal	1 Fourier representation is continuous and a periodic
F	Continuous and periodic signal	2 Fourier representation is discrete and a periodic
G	Discrete and aperiodic signal	3 Fourier representation is continuous and periodic
H	Discrete and periodic signal	4 Fourier representation is discrete and periodic

- (A) E – 3 F – 2 G – 4 H – 1
 (B) E – 1 F – 3 G – 2 H – 4
 (C) E – 1 F – 2 G – 3 H – 4
 (D) E – 2 F – 1 G – 4 H – 3

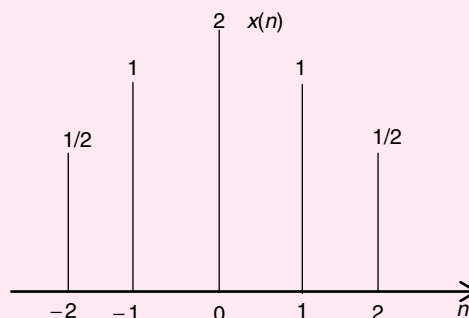
4. A signal $x(n] = \sin(\omega_0 n + \phi)$ is the input to a linear time invariant system having a frequency response $H(e^{j\omega})$. If the output of the system is $Ax(n - n_0)$, then the most general form of $\angle H(e^{j\omega})$ will be [2005]

- (A) $-n_0\omega_0 + \beta$ for any arbitrary real β
 (B) $-n_0\omega_0 + 2\pi k$ for any arbitrary integer k
 (C) $n_0\omega_0 + 2\pi K$ for any arbitrary integer K
 (D) $-n_0\omega_0 + \phi$

Direction for questions 5 and 6:

A sequence $x(n)$ had non-zero values as shown in figure.

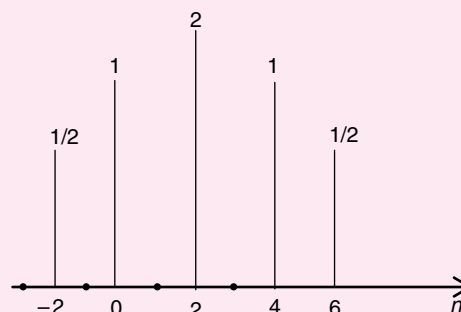
[2005]



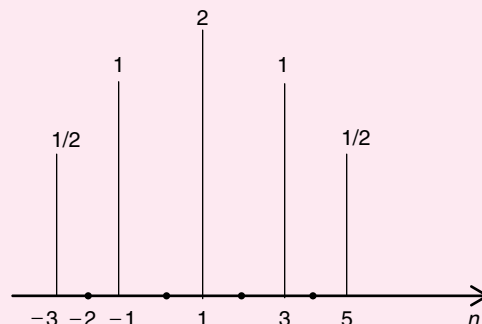
5. The sequence

$$y(n] = \begin{cases} x\left(\frac{n}{2} - 1\right) & \text{for } n \text{ even} \\ 0 & \text{for } n \text{ odd} \end{cases}$$

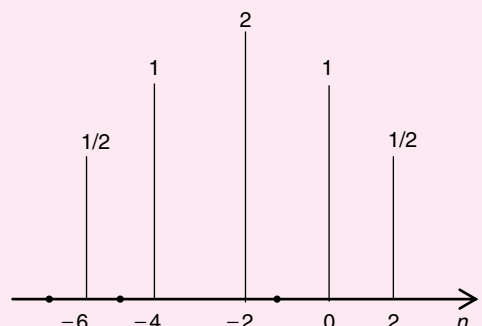
- (A)

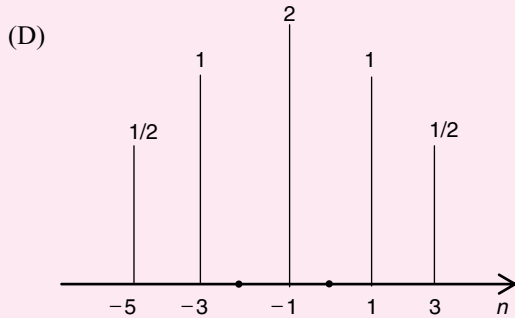


- (B)



- (C)





6. The Fourier transform of $y(2n)$ will be [2005]

(A) $e^{-j2\omega} [\cos 4\omega + 2 \cos 2\omega + 2]$
 (B) $[\cos 2\omega + 2 \cos \omega + 2]$
 (C) $e^{-j\omega} [\cos 2\omega + 2 \cos \omega + 2]$
 (D) $e^{j2\omega} [\cos 2\omega + 2 \cos \omega + 2]$

7. A 5-point sequence $x(n)$ is given as $x[-3] = 1, x[-2] = 1, x[-1] = 0, x[0] = 5, x[1] = 1$. Let $X(e^{j\omega})$ denotes the discrete-time Fourier transform of $x(n)$, the value of $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$ is: [2007]

(A) 5 (B) 10π
 (C) 16π (D) $5 + j10\pi$

8. $\{x(n)\}$ is a real-valued periodic sequence with a period N . $x(n)$ and $X(k)$ form N -point Discrete Fourier transform (DFT) pairs. The DFT $Y(k)$ of the sequence $y(n) = \frac{1}{N} \sum_{r=0}^{N-1} x(r)x(n+r)$ is [2008]

$$y(n) = \frac{1}{N} \sum_{r=0}^{N-1} x(r)x(n+r) \text{ is}$$

(A) $|X(k)|^2$
 (B) $\frac{1}{N} \sum_{r=0}^{N-1} X(r)X^*(k+r)$
 (C) $\frac{1}{N} \sum_{r=0}^{N-1} x(r)X(k+r)$
 (D) 0

9. The 4-point discrete Fourier transform (DFT) of a discrete time sequence $\{1, 0, 2, 3\}$ is [2009]

(A) $[0, -2+2j, 2, -2-2j]$
 (B) $[2, 2+2j, 6, 2-2j]$
 (C) $[6, 1-3j, 2, 1+3j]$
 (D) $[6, -1+3j, 0, -1-3j]$

10. For an N -point FFT algorithm with $N = 2^m$, which one of the following statements is TRUE? [2010]

(A) It is not possible to construct a signal flow graph with both input and output in normal order
 (B) The number of butterflies in the m^{th} stage is N/m

(C) In-place computation requires storage of only $2N$ node data
 (D) Computation of a butterfly requires only one complex multiplication

11. The first six points of the 8-point DFT of a real valued sequence are 5, $1-j3$, 0, $3-j4$, 0, and $3+j4$. The last two points of the DFT are, respectively [2011]

(A) 0, $1-j3$ (B) 0, $1+j3$
 (C) $1+j3$, 5 (D) $1-j3$, 5

12. The DFT of a vector $[a \ b \ c \ d]$ is the vector $[\alpha \ \beta \ \gamma \ \delta]$. Consider the product

$$[p \ q \ r \ s] = [a \ b \ c \ d] \begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}$$

The DFT of the vector $[p \ q \ r \ s]$ is a scaled version of [2013]

(A) $[\alpha^2 \ \beta^2 \ \gamma^2 \ \delta^2]$
 (B) $[\sqrt{\alpha} \ \sqrt{\beta} \ \sqrt{\gamma} \ \sqrt{\delta}]$
 (C) $[\alpha + \beta \ \beta + \delta \ \delta + \gamma \ \gamma + \alpha]$
 (D) $[\alpha \ \beta \ \gamma \ \delta]$

13. Consider a discrete-time periodic signal $x(n) = \sin\left(\frac{\pi n}{5}\right)$. Let a_k be the complex Fourier series coefficients of $x(n)$. The coefficients $\{a_k\}$ are non-zero when $k = Bm \pm 1$, where m is any integer. The value of B is [2014]

14. A Fourier transform pair is given by

$$\left(\frac{2}{3}\right)^n u(n+3) \stackrel{FT}{\Leftrightarrow} \frac{A e^{-j6\pi f}}{1 - \left(\frac{2}{3}\right) e^{-j2\pi f}}$$

where $u(n)$ denotes the unit step sequence. The values of A is [2014]

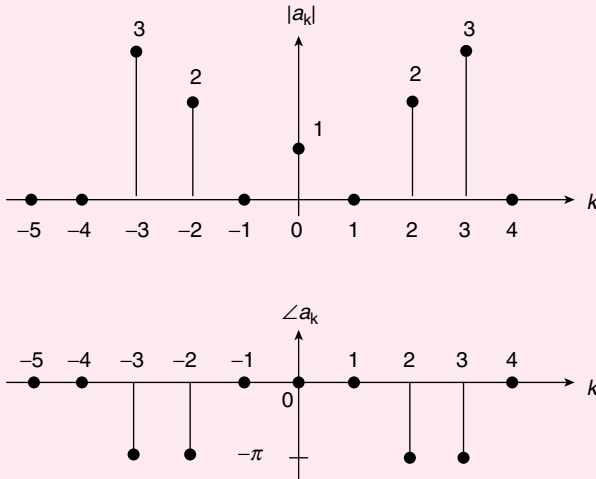
15. The N -point DFT X of a sequence $x(n)$, $0 \leq n \leq N-1$ is given by

$$X(k) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}nk}, \quad 0 \leq k \leq N-1$$

Denote this relation as $X = \text{DFT}(x)$. For $N = 4$, which one of the following sequences satisfies $\text{DFT}(\text{DFT}(x)) = x$? [2014]

(A) $x = [1 \ 2 \ 3 \ 4]$ (B) $x = [1 \ 2 \ 3 \ 2]$
 (C) $x = [1 \ 3 \ 2 \ 2]$ (D) $x = [1 \ 2 \ 2 \ 3]$

16. The magnitude and phase of the complex Fourier series coefficient a_k of a periodic signal $x(t)$ are shown in the figure. Choose the correct statement from the four choices given. Notation C is the set of complex numbers, R is the set of purely real numbers, and P is the set of purely imaginary numbers. [2015]



- (A) $x(t) \in R$
 (B) $x(t) \in P$
 (C) $x(t) \in (C - R)$
 (D) the information given is not sufficient to draw any conclusion about $x(t)$
17. Consider a four-point moving average filter defined by the equation $y[n] = \sum_{i=0}^3 \alpha_i x[n-i]$. The condition on the filter coefficients that results in a null at zero frequency is [2015]
- (A) $\alpha_1 = \alpha_2 = 0; \alpha_0 = -\alpha_3$
 (B) $\alpha_1 = \alpha_2 = 1; \alpha_0 = -\alpha_3$
 (C) $\alpha_0 = \alpha_3 = 0; \alpha_1 = \alpha_2$
 (D) $\alpha_1 = \alpha_2 = 0; \alpha_0 = \alpha_3$
18. Consider the signal $x[n] = 6\delta[n+2] + 3\delta[n+1] + 8\delta[n] + 7\delta[n-1] + 4\delta[n-2]$. If $x(e^{j\omega})$ is the discrete-time Fourier transform of $x[n]$, then $\frac{1}{\pi} \int_{-\pi}^{\pi} x(e^{j\omega}) \sin^2(2\omega) d\omega$ is equal to _____. [2016]

19. A continuous time filter with transfer function $H(s) = \frac{2s+6}{s^2+6s+8}$ is converted to a discrete time filter with transfer function $G(z) = \frac{2z^2 - 0.5032z}{z^2 - 0.5032z + k}$ so that the impulse response of the continuous time filter, sampled at 2Hz, is identical at the sampling instants to the impulse response of the discrete time filter. The value of k is _____. [2016]

20. The Discrete Fourier Transform (DFT) of the 4 point sequence [2016]

$$x[n] = \{x[0], x[2], x[3]\} = \{3, 2, 3, 4\}$$

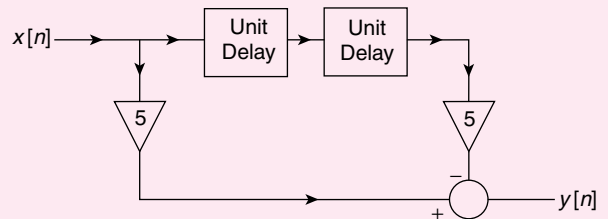
$$x[k] = \{x[0], x[1], x[2], x[3]\} = \{12, 2j, 0, -2j\}$$

If $x_1[k]$ is the DFT of the 12 point sequence $x_1[n] =$

$\{3, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0\}$ the value of $\left| \frac{x_1[8]}{x_1[1]} \right|$ is _____. [2016]

21. A continuous time speech signal $x_a(t)$ is sampled at a rate of 8 kHz and the samples are subsequently grouped in blocks, each of size N . The DFT of each block is to be computed in real time using the radix-2 decimation in frequency FFT algorithm. If the processor performs all operations sequentially, and takes 20 ms for computing each complex multiplication (including multiplications by 1 and -1) and the time required for addition/subtraction is negligible, then the maximum value of N is _____. [2016]

22. The direct form structure of an FIR (finite impulse response) filter is shown in the figure:



The filter can be used to approximate a _____. [2016]

- (A) low pass filter (B) high pass filter
 (C) band pass filter (D) band stop filter

ANSWER KEYS**EXERCISES****Practice Problems 1**

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|------|------|------|------|-------|
| 1. C | 2. A | 3. C | 4. D | 5. A | 6. C | 7. D | 8. C | 9. C | 10. A |
| 11. A | 12. B | 13. D | 14. A | 15. B | | | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|------|------|------|------|-------|
| 1. A | 2. A | 3. A | 4. B | 5. C | 6. D | 7. A | 8. A | 9. B | 10. A |
| 11. C | 12. B | 13. A | 14. D | 15. A | | | | | |

Previous Years' Questions

- | | | | | | | | | | |
|-----------|-------|-------------------|-------|-----------|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. C | 4. B | 5. A | 6. C | 7. B | 8. A | 9. D | 10. B |
| 11. B | 12. A | 13. 9.99 to 10.01 | | 14. 3.375 | 15. B | 16. A | 17. A | 18. 8 | |
| 19. 0.049 | 20. 6 | 21. 4096 | 22. C | | | | | | |

SIGNALS AND SYSTEMS

Time: 60 Minutes

Direction for questions 1 to 30: Select the correct alternative from the given choices.

1. The function $f(t)$ having Laplace transform at a signal

$$y(t) \text{ is } Y(s) = \frac{1}{s(s-6)}, \text{ then its final value is}$$

- (A) -6 (B) 0
(C) 6 (D) Unbounded

2. Laplace transform at $e^{(3b+2)t+8}$ exists if _____

- (A) $\operatorname{Re}(s) > 3b + 2$ (B) $\operatorname{Re}(s) < 2$
(C) $\operatorname{Re}(s) > 3b + 8$ (D) $\operatorname{Re}(s) > 3b + 10$

3. If $F(s) = \mathcal{L}[f(t)] = \frac{K}{(s+6)(s^2+8)}$ then $\lim_{t \rightarrow \infty} f(t)$ is given by _____

- (A) $K/8$ (B) Zero
(C) infinite (D) Undefined

4. Function $f(t)$ having Laplace transform

$$F(s) = \frac{s}{s^2 + \omega_o^2} \quad \operatorname{Re}(s) > 0$$

The final value of $f(t)$ would be

- (A) 0 (B) ∞
(C) 1 (D) $-1 \leq f(\infty) \leq 1$

5. If $|b| < |a|$, then the z -transform $X(z)$ of the signal $x(n) = a^n u(n) + b^n u(-n-1)$ is

- (A) Zero (B) Does not exist
(C) $\frac{1}{1-az^{-1}} - \frac{1}{1-bz^{-1}}$ (D) $\frac{1}{1-az^{-1}} + \frac{1}{1-bz^{-1}}$

6. The step response of the causal system is described by the difference equation $y(n) = y(n-1) + x(n)$, so $y(n)$ is

- (A) Causal and stable
(B) Anticausal and stable
(C) Causal and unstable
(D) Anticausal and unstable

7. An LTI system described by difference equation $y(n) - y(n-1) + 1/4 y(n-2)$

$$= x(n) - \frac{3}{4} x(n-1) - \frac{1}{8} x(n-2)$$

The LTI inverse system would be

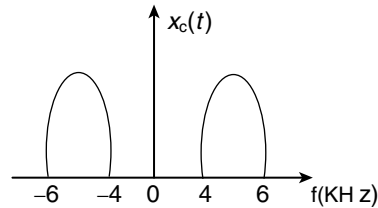
- (A) Stable and causal (B) Unstable and causal
(C) Stable and anticausal (D) None of these

8. The final value of the signal

$$Y(z) = \frac{z^{-1}(1-z^{-4})}{2(1-z^{-1})^2}$$

- (A) 0 (B) 2 (C) 1 (D) ∞

9. For a band pass signal $x_c(t)$ is shown in the following figure. The minimum sampling rate required for this signal to prevent loss of information is



- (A) 4 KHz (B) 12 KHz
(C) 20 KHz (D) 2 KHz

10. A continuous time signal for the duration of 5.5 sec is uniformly sampled without aliasing and generating a finite length sequence containing 3,000 samples. What is the highest frequency component that could be present in continuous time signal?

- (A) 277.5 Hz (B) 3,000 Hz
(C) 200 Hz (D) 500 Hz

11. A band-limited signal with a maximum frequency of 10 KHz is to be sampled. According to sampling theorem, the sampling frequency which is not valid is

- (A) 10 KHz (B) 20 KHz
(C) 30 KHz (D) 25 KHz

12. A signal $x(t)$ that is band limited to B Hz, and whose samples are

$$x(0) = 1 \text{ and } x(\pm T) = x(\pm 2T) = x(\pm 3T) = \dots = 0$$

Where the sampling interval T is the Nyquist interval

$$\text{for } x(t), \text{ that is } T = \frac{1}{2B}$$

So signal $x(t)$ is

- (A) $\delta(t)$ (B) $\sin(2\pi Bt)$
(C) $\operatorname{sinc}(2\pi Bt)$ (D) $\operatorname{rect}(Bt)$

13. A 4 KHz sinusoidal signal is ideally sampled at 6 KHz, and the sampled signal is passed through an ideal low-pass filter with cut-off frequency 3 KHz. The output signal has the frequency.

- (A) Zero (B) 2 KHz
(C) 3 KHz (D) 1 KHz

14. The condition on the sampling interval T_s so that $x(t)$ is uniquely represented by discrete time sequence $x[n] = x(nT_s)$

$$x(t) = \cos(4\pi t) \frac{\sin 2\pi t}{\pi t} + 8 \sin(8\pi t) \frac{\sin(\pi t)}{\pi t}$$

- (A) $T_s < \frac{1}{6}$ (B) $T_s < \frac{1}{9}$

- (C) $T_s < \frac{1}{4}$ (D) $T_s < \frac{1}{2}$

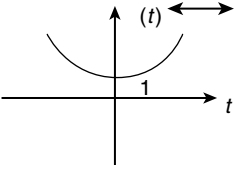
15. z transform for $\cos \beta n$ is, (where $\beta = \frac{2\pi}{3}$)

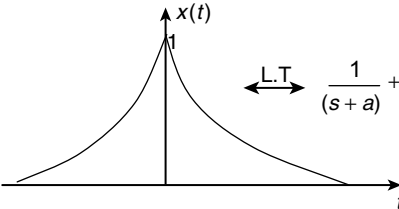
- (A) $\pi \sum_{k=-\infty}^{+\infty} [\delta(z - \beta - 2\pi K) + \delta(z + \beta - 2\pi K)]$
 (B) $\pi [\delta(z - \beta) + \delta(z + \beta)]$
 (C) $j\pi \sum_{k=-\infty}^{+\infty} [\delta(z - \beta - 2\pi K) - \delta(z + \beta - 2\pi K)]$
 (D) does not exist

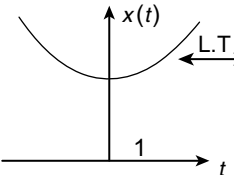
16. If $x(t)$ is right-sided signal, then the ROC of the signal is in the form as _____

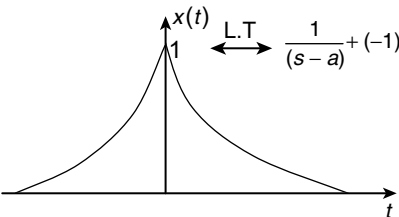
- (A) $\text{Re}(s) > \sigma_1$ (B) $\text{Re}(s) < \sigma_2$
 (C) $\sigma_1 < \text{Re}(s) < \sigma_2$ (D) Entire s plane

17. For $a < 0$, the signal $x(t) = e^{-a|t|}$ and Laplace transform of $x(t)$ can be expressed as

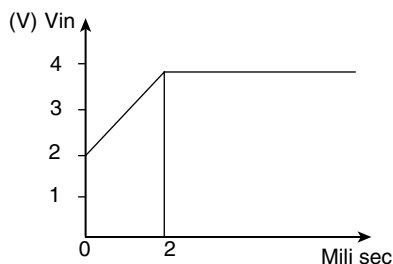
- (A)  $\xleftrightarrow{\text{L.T.}} \frac{1}{(s-a)} + (-1) \frac{1}{(s+a)}$

- (B)  $\xleftrightarrow{\text{L.T.}} \frac{1}{(s+a)} + \frac{(-1)}{(s-a)}$

- (C)  $\xleftrightarrow{\text{L.T.}} \frac{1}{(s+a)} + \frac{(-1)}{(s-a)}$

- (D)  $\xleftrightarrow{\text{L.T.}} \frac{1}{(s-a)} + (-1) \frac{1}{(s+a)}$

18. The output of the differentiator whose input V_{in} is given as _____



- (A) $2 \delta(t) - 1000r(t) + 1000r(t-2)$
 (B) $2 \delta(t) + 1000 u(t) - 1000 u(t-2)$
 (C) $2 r(t) + 4 u(t-2) + 2 u(t)$
 (D) $2 u(t) + 1000 r(t) - 1000r(t-2)$

19. If a system whose transfer function

$$H(s) = \frac{s+0.2}{s+4}$$

Then the system response $y(t)$ for $x(t) \cos(9t - 30^\circ)$ is

- (A) $0.914 \cos(9t + 22.69^\circ)$
 (B) $1.09 \cos(9t + 22.69^\circ)$
 (C) $0.914 \cos(9t - 7.310^\circ)$
 (D) $1.09 \cos(9t - 7.310^\circ)$

20. If $x[n] = \left(\frac{1}{7}\right)^{|n|} - \frac{1}{3} u(n)$, then the ROC of its z transform in z plane will be

- (A) $|z| > \frac{1}{7}$ (B) $\frac{1}{7} < |z| < 7$
 (C) $\frac{1}{3} < |z| < 7$ (D) $\frac{1}{7} < |z| < \frac{1}{3}$

21. z transform $X(z)$ at a sequence $x[n]$ is given by $X(z) = \frac{0.6}{\left(1 - \frac{20}{5} z^{-1}\right)}$. It is given that the ROC of $X(z)$ excludes

the unit circle. The value of $X[z]$ is

- (A) 0 (B) 9.6 (C) 0.6 (D) -0.6

22. Transfer function for a system is given as

$$H(Z) = \frac{5}{\left(1 - \frac{1}{3} z^{-1}\right)} + \frac{3}{(1 - 4z^{-1})}$$

Match the type of system with their corresponding ROCs.

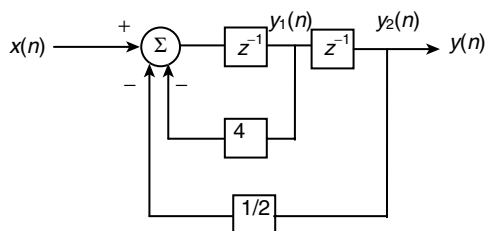
Type of system	ROCs
(1) Stable system	(I) $ Z > 4$
(2) Causal system	(II) $ Z < \frac{1}{3}$
(3) Anticausal system	(III) $\frac{1}{3} < Z < 4$

- (A) 1 - II, 2 - III, 3 - I
 (B) 1 - III, 2 - I, 3 - II
 (C) 1 - I, 2 - II, 3 - III
 (D) 1 - II, 2 - I, 3 - III

23. The time signal $x[n]$, whose z transform $X(z)$ has poles at $z = 1/4$ and $z = -1$. Also given that $x[2] = 2$ & $x(-1) = 2$ and ROC includes the point $z = 1/2$. Then, the time signal $x(n)$ is _____

- (A) $8\left(\frac{1}{4}\right)^n u(n) - (-1)^n u(-n-1)$
 (B) $8\left(\frac{1}{4}\right)^n u(n) - 2(-1)^n u(-n-1)$
 (C) $32\left(\frac{1}{4}\right)^n u(n) - 2(-1)^n u(-n-1)$
 (D) $2(-1)^n u(n) - 16\left(\frac{1}{4}\right)^n u(-n-1)$

24. An LTI system is given below for $n < 0$. Assume that $x(n]$, $y_1(n]$, and $y_2(n]$ are all zero.



The transfer function

$$H(z) = \frac{Y_2(z)}{X(z)}$$

- (A) $\frac{2z^{-2}}{1 + 2z^{-1} + \frac{1}{2}z^{-2}}$ (B) $\frac{z^{-2}}{1 + 4z^{-1} + \frac{1}{2}z^{-2}}$
 (C) $\frac{z^{-2}}{1 + 4z^{-2} + \frac{1}{2}z^{-1}}$ (D) None of these
25. The inverse Z transform of $X(z) = \frac{2 + Z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)}$ with ROC $|Z| > \frac{1}{2}$
- (A) $2\delta(n) + 2\delta(n-1) + \delta(n-2) + \frac{1}{2}\delta(n-3) + \dots$
 (B) $-2\delta(n) - 8\delta(n+1) - 16\delta(n+2) - 32\delta(n+3) + \dots$
 (C) $-4\delta(n+1) - \delta(n+1) - 16\delta(n+3) - 32\delta(n+4) + \dots$
 (D) $\delta(n-1) + \frac{1}{2}\delta(n-2) + \frac{1}{4}\delta(n-3) + \frac{1}{8}\delta(n-4) + \dots$

26. An analogue signal

$$x_a(t) = 3\cos 8000\pi t + 8\sin 6000\pi t + 100\cos 10,000\pi t$$

is

Sampled at a rate $F_s = 3000$ samples/sec

The discrete time signal obtained after sampling is

- (A) $3\cos 8\pi t + 8\sin 6\pi t + 100\cos 10\pi t$
 (B) $3\cos 8000\pi t + 8\sin 6000\pi t$
 (C) $103\cos \frac{2\pi}{3}n + 8\sin \frac{2\pi}{3}n$
 (D) $103\cos 2\pi\left(\frac{1}{3}\right)n$

Direction for questions 27 and 28:

27. Transfer function for a causal system is given as

$$H(z) = \frac{z^{-2} + z^{-1}}{1 - Pz^{-1} - Pz^{-2}}$$

The range of values of P so that system is stable

- (A) $\left(-1, \frac{1}{2}\right)$ (B) $\left(-\frac{1}{2}, 1\right)$
 (C) $\left(\frac{1}{2}, 2\right)$ (D) $\left(-2, \frac{1}{2}\right)$

28. For which value of P , the system will behave as a notch filter.

- (A) -1 (B) $-1/2$ (C) $1/2$ (D) 2

Direction for questions 29 and 30:

Inverse Laplace transform of $\frac{2s+4}{s^2+4s+3}$

29. When $\text{Re}(s) < -3$
 (A) $e^{-3t}u(t) - e^{-t}u(-t)$ (B) $-e^{-3t}u(-t) + e^{-t}u(t)$
 (C) $-e^{-3t}u(-t) - e^{-t}u(-t)$ (D) $e^{-3t}u(t) + e^{-t}u(t)$
30. When $-3 < \text{Re}(s) < -1$
 (A) $-e^{-3t}u(-t) + e^{-t}u(t)$ (B) $e^{-3t}u(t) - e^{-t}u(-t)$
 (C) $-e^{-3t}u(t) + e^{-t}u(-t)$ (D) $e^{-3t}u(-t) - e^{-t}u(t)$

ANSWER KEYS

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|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. D | 2. A | 3. D | 4. D | 5. B | 6. C | 7. A | 8. B | 9. A | 10. A |
| 11. A | 12. C | 13. B | 14. B | 15. D | 16. A | 17. A | 18. B | 19. C | 20. C |
| 21. B | 22. B | 23. C | 24. B | 25. A | 26. D | 27. A | 28. A | 29. C | 30. B |